ROLL NO: _____

# MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

## END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. EE ENGINEERING — SEMESTER 4

**Time Allowed: 3 Hours** **Maximum Marks: 70**

*NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.*
*2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].*
*3. Assume suitable missing data, if any, and state it clearly.*
*4. Use of scientific non-programmable calculator is permitted.*

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**--- UNIT I ---**

**Q.1** Compare Open-loop and Closed-loop control systems. Find Transfer Function using Block Diagram Reduction. [14 Marks]

**[ OR ]**

**Q.1 (Alt)** State Mason's Gain Formula and evaluate transfer function from Signal Flow Graph. [14 Marks]

**--- UNIT II ---**

**Q.2** Derive Time Response of second-order system to unit step input. Define Rise time, Peak time, Overshoot. [14 Marks]

**[ OR ]**

**Q.2 (Alt)** Explain Steady-State Error and Error Constants (Kp, Kv, Ka) for Type-0, Type-1, Type-2 systems. [14 Marks]

**--- UNIT III ---**

**Q.3** State Routh-Hurwitz Stability Criterion. Determine stability of characteristic equation with special cases. [14 Marks]

**[ OR ]**

**Q.3 (Alt)** Explain Root Locus technique: Rules for plotting Root Locus, Asymptotes, Breakaway points, and Angle of departure. [14 Marks]

**--- UNIT IV ---**

**Q.4** Explain Frequency Response Analysis: Draw Bode Plot. Determine Gain Margin and Phase Margin. [14 Marks]

**[ OR ]**

**Q.4 (Alt)** Explain Polar Plots and Nyquist Stability Criterion. Determine closed-loop stability. [14 Marks]

**--- UNIT V ---**

**Q.5** Explain State Variable Analysis: State Transition Matrix properties and solution of state equation. [14 Marks]

**[ OR ]**

**Q.5 (Alt)** Explain Controllability and Observability using Kalman's test. Describe P, PI, PID controllers. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****